

Respiration

Question Paper 1

Level	IGCSE
Subject	Biology
Exam Board	CIE
Topic	Respiration
Paper Type	(Extended) Theory Paper
Booklet	Question Paper 1

Time Allowed: 34 minutes

Score: /28

Percentage: /100

- 1 Researchers designed an investigation to find the effect of increasing levels of exercise on two groups of people.

The first group of people were trained cyclists and the second group were untrained cyclists.

The researchers asked all the people to cycle at four levels of effort: 30%, 45%, 60% and 75% of their maximum cycle speed.

They cycled for eight minutes at each level of effort.

- (a) The researchers predicted that the pulse rate of all the cyclists would increase during exercise.

Explain this prediction.

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Fig. 3.1 shows the average concentration of lactic acid in the blood of the trained cyclists and untrained cyclists in the investigation.

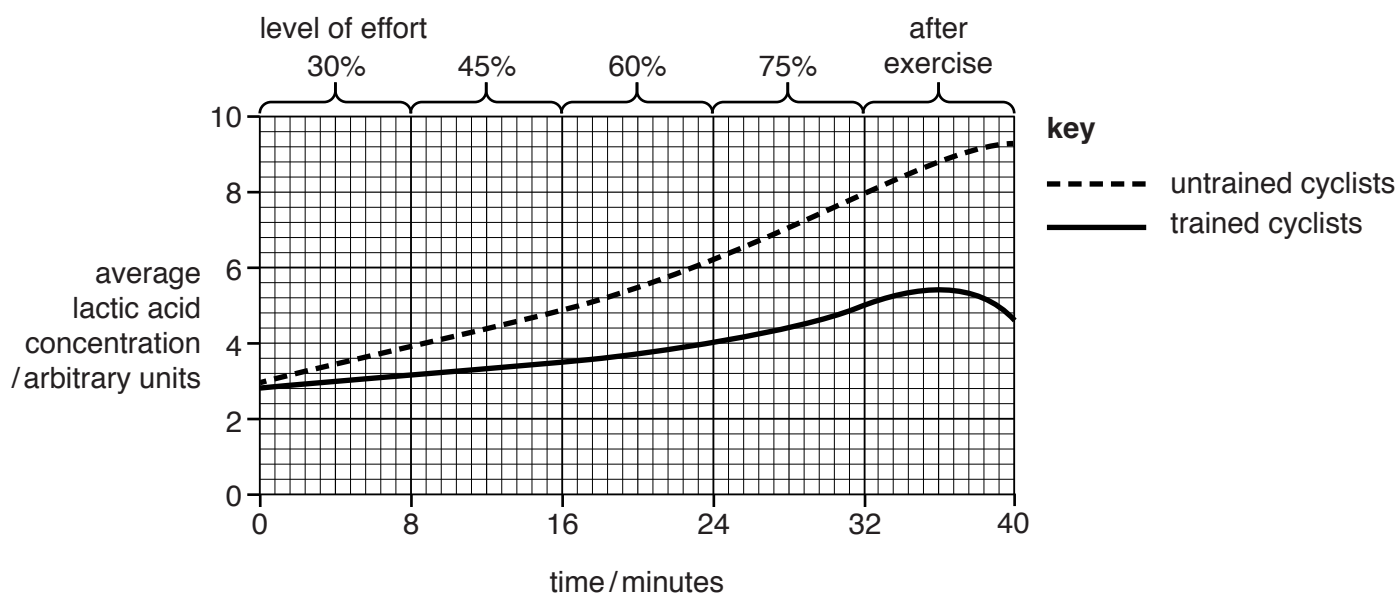


Fig. 3.1

- (b) Describe the effect of the increasing levels of effort on the average lactic acid concentration in the blood of the **untrained** cyclists.

You should use data from Fig. 3.1 in your answer.

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- (c) Lactic acid is produced in the muscles during anaerobic respiration.

- (i) Define the term *anaerobic respiration*.

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- (ii) Describe how the lactic acid produced in muscle cells enters the blood.

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- (iii) Name the component of the blood that transports lactic acid.

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- (d) Explain why the lactic acid concentration in the blood in trained cyclists is different from the untrained cyclists eight minutes **after** the exercise.

You should use data from Fig. 3.1 in your answer.

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[Total: 13]

- 2 (a) Define the term *respiration*.

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- (b) A rowing machine is a piece of apparatus that is used in many fitness centres.

Fig. 4.1 shows a man training on a rowing machine. The man in the photograph has his arms extended during the rowing stroke as shown in Fig. 4.2.

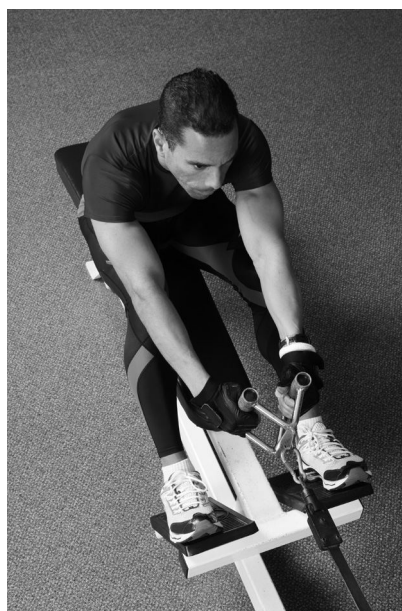


Fig. 4.1

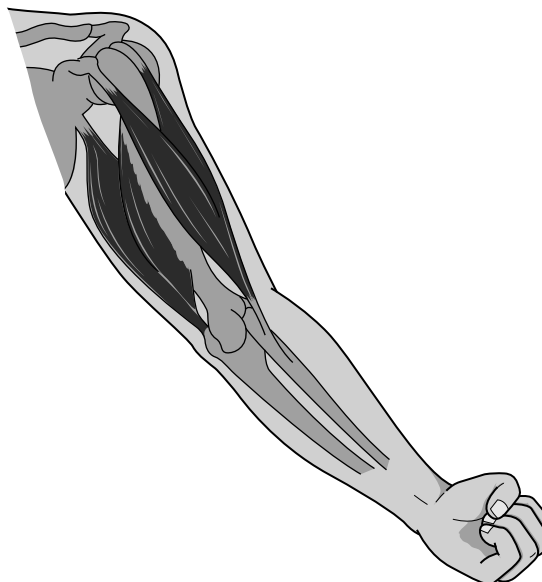


Fig. 4.2

Use Fig. 4.2 to describe how the hand is moved closer to the chest during the rowing stroke.

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- (c) The man has an intense workout on the rowing machine.

Fig. 4.3 shows his oxygen uptake before and during the exercise.

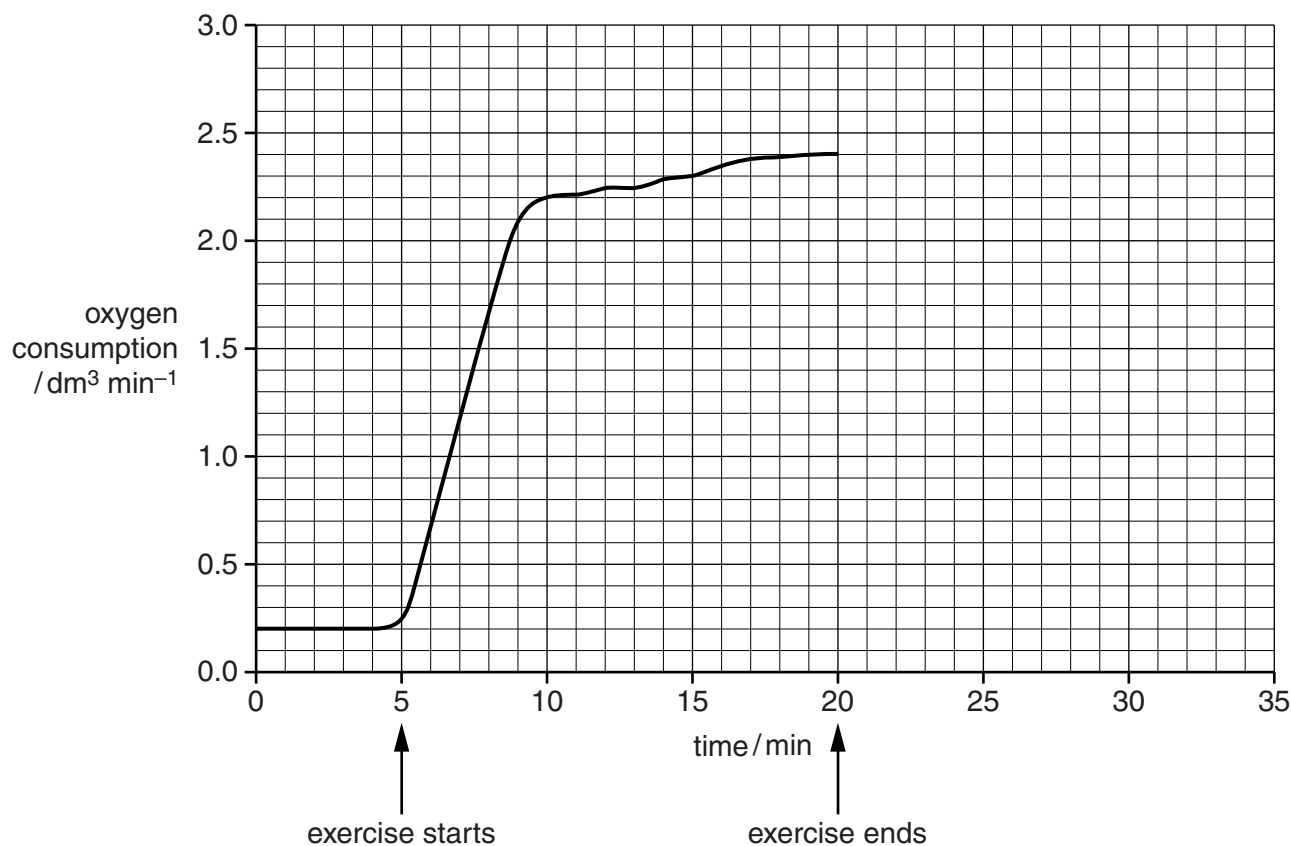


Fig. 4.3

- (i) Explain why there is a steep increase in the man's oxygen consumption at the start of the exercise.

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- (ii) It took 10 minutes after the man had stopped rowing for his oxygen consumption to decrease to its resting value.

On Fig. 4.3 draw a line between 20 minutes and 35 minutes to show the change in oxygen consumption after exercise has stopped. [2]

- (iii) Explain why the man's oxygen consumption did not return to the resting value **immediately** after exercise.

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[Total: 15]